

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost &gt; 25/episode)



- SafeOT-typed (ours)
 ▼ SDAC (multi-constraint SOTA)
 ⋯ SAC-Pen coefficient sweep
- FDPI (constraint-based)
 + SRCPO (multi-constraint SOTA)
 --- FDPI  $p_f$  sweep
- ◆ SAC-Lagrangian (penalty-based)
 — typed Pareto frontier (soft  $\leq 25$ )
 --- SAC-Lag cost-limit sweep
- ▲ SAC + fixed penalty (penalty-based)